极限

数列极限

$$
\begin{gathered}
\text{若}\forall\epsilon>0,\exists N>0,\text{当}n>N\text{时},\\
|a_n-A|<\epsilon\\
\lim_{n\to\infty}a_n=A\text{或}a_n\to A(n\to\infty)
\end{gathered}
$$

$$
a_n=\frac{n+1}{2n},\lim_{n\to\infty}a_n=\frac{1}{2},\frac{n+1}{2n}\neq\frac{1}{2}
$$

$$
\begin{gathered}
\text{证明：若}\lim_{n\to\infty}a_n=A,\text{则}\lim_{n\to\infty}|a_n|=|A|,\text{反之不对.}\\
\Rightarrow\\
\because\lim_{n\to\infty}a_n=A\\
\therefore\forall\epsilon>0,\exists N>0,\text{当}n>N\text{时},\\
|a_n-A|<\epsilon\\
\because||a_n|-|A||\leq|a_n-A|\\
\therefore\forall\epsilon>0,\exists N>0,\text{当}n>N\text{时}\\
||a_n|-|A||<\epsilon\\
\therefore\lim_{n\to\infty}|a_n|=|A|\\
\not\Leftarrow\\
a_n=(-1)^n,\lim_{n\to\infty}a_n\text{不存在},\text{但}\lim_{n\to\infty}|a_n|=1
\end{gathered}
$$

epsilon-delta

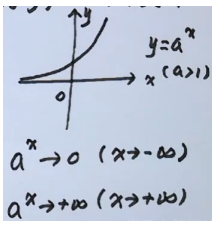
$$
\begin{gathered}
f(x)\text{在}x=a\text{的去心邻域内有定义}\\
\text{若}\forall\epsilon>0,\exists\delta>0,\text{当}0<|x-a|<\delta,\text{有}\\
|f(x)-A|<\epsilon\\
\lim_{x\to a}f(x)=A\text{或}f(x)\to A(x\to a)
\end{gathered}
$$

$$
\begin{aligned}
&1.\quad x\to a:\begin{cases}x\neq a\\x\to a^-,x\to a^+\end{cases}\\
&2.\quad\lim_{x\to a}f(x)\text{与}f(a)\text{无关}
\end{aligned}
$$

$$
\begin{gathered}
f(x)=\frac{x^2+x-2}{x^2-1}(x\neq\pm1)\\
\lim_{x\to1}f(x)=\lim_{x\to1}\frac{(x-1)(x+2)}{(x-1)(x+1)}=\lim_{x\to1}\frac{x+2}{x+1}=\frac{3}{2},\text{而}f(1)\text{不存在}
\end{gathered}
$$

$$
\begin{aligned}
&3.\ 若\forall\epsilon>0,\exists\delta_1>0,当x\in(a-\delta_1,a)时\\
&|f(x)-A|<\epsilon\\
&\lim_{x\to a^-}f(x)=A或f(a-0)=A\\
&若\forall\epsilon>0,\exists\delta_2>0,当x\in(a,a+\delta_2)时\\
&|f(x)-B|<\epsilon\\
&\lim_{x\to a^+}f(x)=B或f(a+0)=B\\
&\lim_{x\to a}f(x)\exists\Leftrightarrow f(a-0),f(a+0)\exists且等
\end{aligned}
$$

$$f(x)含a^{\frac{?}{x-b}}或a^{\frac{?}{b-x}}当x\to b时分左右极限$$



$$设f(x)=\frac{1-2^{\frac{1}{x-1}}}{1+2^{\frac{1}{x-1}}},判断\lim_{x\to1}f(x)是否存在.$$

$$x\to1^-\Rightarrow\frac{1}{x-1}\to-\infty\Rightarrow2^{\frac{1}{x-1}}\to0$$

$$f(1-0)=1$$

$$x\to1^+\Rightarrow\frac{1}{x-1}\to+\infty\Rightarrow2^{\frac{1}{x-1}}\to+\infty$$

$$f(1+0)=-1$$

$$\because f(1-0)\neq f(1+0),\therefore f(x)不存在$$

epsilon-x

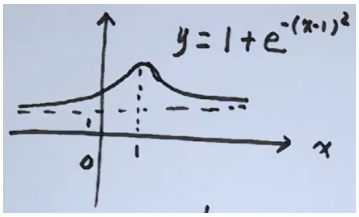
case1

$$
\begin{aligned}
&若\forall\epsilon>0,\exists X>0,当x>X时,\\
&|f(x)-A|<\epsilon\\
&\lim_{x\to+\infty}f(x)=A
\end{aligned}
$$

case2

$$
\begin{aligned}
&若\forall\epsilon>0,\exists X>0,当x<-X时,\\
&|f(x)-A|<\epsilon\\
&\lim_{x\to-\infty}f(x)=A
\end{aligned}
$$

case3



$$\text{若}\forall\epsilon>0,\exists X>0,\text{当}|x|>X\text{时},$$
$$|f(x)-A|<\epsilon$$
$$\lim_{x\to\infty}f(x)=A$$

一般性质

唯一性

$$\text{极限存在必唯一}$$

$$\text{证：设}\lim_{x\to a}f(x)=A,\lim_{x\to a}f(x)=B$$
$$\text{若}A>B:\text{取}\epsilon=\frac{A-B}{2}>0,$$
$$\because\lim_{x\to a}f(x)=A,\therefore\exists\delta_1>0,\text{当}0<|x-a|<\delta_1\text{时},$$
$$|f(x)-A|<\frac{A-B}{2}$$
$$\text{即}\frac{A+B}{2}<f(x)<\frac{3A-B}{2}(*)$$
$$\because\lim_{x\to a}f(x)=B,\therefore\exists\delta_2>0,\text{当}0<|x-a|<\delta_2\text{时},$$
$$|f(x)-B|<\frac{A-B}{2}$$
$$\text{即}\frac{3B-A}{2}<f(x)<\frac{A+B}{2}(**)$$
$$\text{取}\delta=\min\{\delta_1,\delta_2\},\text{当}0<|x-a|<\delta\text{时},(*)(**)\text{皆对矛盾}$$
$$\therefore A>B\text{不对}$$
$$\text{若}A<B:\text{同理不对}$$
$$\therefore A=B$$

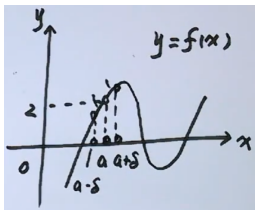
有界性

$$\text{若}\lim_{n\to\infty}a_n=A\Rightarrow\exists M>0,\forall n,|a_n|\le M$$
$$\not\Leftarrow$$

$$\text{证}:\Rightarrow\text{取}\epsilon=1$$
$$\because\lim_{n\to\infty}a_n=A,\therefore\lim_{n\to\infty}|a_n|=|A|$$
$$\therefore\exists N>0,\text{当}n>N\text{时},\text{有}$$
$$||a_n|-|A||<1\Rightarrow|a_n|<1+|A|$$
$$\text{取}M=\max\{|a_1|,|a_2|,\dots,|a_N|,1+|A|\}$$
$$\therefore\forall n,|a_n|\le M$$
$$\not\Leftarrow a_n=2+(-1)^n,|a_n|\le 3,\text{但}\lim_{n\to\infty}a_n\text{不存在}$$

保号性

$$\text{若}\lim_{x\to a}f(x)=A\begin{cases}>0\\<0\end{cases},\text{则}$$
$$\exists\delta>0,\text{当}0<|x-a|<\delta\text{时},$$
$$f(x)\begin{cases}>0\\<0\end{cases}$$



$$\lim_{x\to a} f(x) = 2 > 0$$
$$\text{当}0 < |x-a| < \delta\text{时}$$
$$f(x) > 0$$

$$A > 0$$
$$\text{取}\epsilon = \frac{A}{2}$$
$$\because \lim_{x\to a} f(x) = A$$
$$\therefore \exists\delta > 0, \text{当}0 < |x-a| < \delta\text{时}$$
$$|f(x) - A| < \frac{A}{2}$$
$$\Rightarrow f(x) > \frac{A}{2} > 0$$
$$A < 0$$
$$\text{取}\epsilon = -\frac{A}{2}$$
$$\because \lim_{x\to a} f(x) = A$$
$$\therefore \exists\delta > 0, \text{当}0 < |x-a| < \delta\text{时}$$
$$|f(x) - A| < -\frac{A}{2}$$
$$\Rightarrow f(x) < \frac{A}{2} < 0$$

$$\text{设}\lim_{x\to 1}\frac{f(x)-3}{\ln^2 x} = -2\text{且}f(1) = 3, \text{讨论}x = 1\text{是否为函数}f(x)\text{的极值点.}$$
$$\because \lim_{x\to 1}\frac{f(x)-3}{\ln^2 x} = -2 < 0$$
$$\therefore \exists\delta > 0, \text{当}0 < |x-1| < \delta\text{时}, \frac{f(x)-3}{\ln^2 x} < 0$$
$$\because \ln^2 x > 0, \therefore f(x) - 3 < 0$$
$$\text{即当}0 < |x-1| < \delta\text{时}, f(x) < f(1)$$
$$\therefore x = 1\text{为}f(x)\text{的极大点}$$

$$\text{设}f'(0) = 0, \text{又}\lim_{x\to 0}\frac{f'(x)}{x^3} = -2, \text{讨论}x = 0\text{是否为}f(x)\text{的极值点.}$$

$$\because \lim_{x\to 0}\frac{f'(x)}{x^3}=-2<0,$$

$$\therefore \exists\delta>0,\text{当}0<|x|<\delta\text{时},\frac{f'(x)}{x^3}<0$$

$$\begin{cases}f'(x)>0,x\in(-\delta,0)\\f'(x)<0,x\in(0,\delta)\end{cases}\Rightarrow x=0\text{为}f(x)\text{的极大点}$$

运算性质

四则

1. 若 $\lim f(x),\lim g(x)\exists\Rightarrow\lim[f(x)\pm g(x)]=\lim f(x)\pm\lim g(x)$
2. 若 $\lim f(x),\lim g(x)\exists\Rightarrow\lim f(x)g(x)=\lim f(x)\lim g(x)$
3. 若 $\lim f(x)\exists,\lim g(x)\exists\neq 0\Rightarrow\lim\dfrac{f(x)}{g(x)}=\dfrac{\lim f(x)}{\lim g(x)}$

复合

1. $\lim\limits_{u\to a}f(u)=A,\lim\limits_{x\to x_0}\Phi(x)=a\Rightarrow$
   $\lim\limits_{x\to x_0}f[\Phi(x)]=A$
2. $\lim\limits_{u\to a}f(u)=f(a),\lim\limits_{x\to x_0}\Phi(x)=a\Rightarrow$
   $\lim\limits_{x\to x_0}f[\Phi(x)]=f[\lim\limits_{x\to x_0}\Phi(x)]=f(a)$